



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2023 Year 11 Course Assessment Task 2

Wednesday, 21st June 2023

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Answer each question in a separate writing booklet. Extra writing booklets are available.
- Write using a blue or a black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- A NESA Reference Sheet is provided.

Class (please ✓)

- ☐ 11MAX.1 – Ms Ham
- ☐ 11MAX.2 – Miss Chai
- ☐ 11MAX.3 – Miss J. Kim / Mr Lam
- ☐ 11MAX.4 – Mr Ho
- ☐ 11MAX.5 – Mr Sekaran
- ☐ 11MAX.6 – Miss Lee

NESA STUDENT #

Marker's use only.

QUESTION	1	2	3	4	5	6	Total	%
MARKS	$\overline{7}$	$\overline{6}$	$\overline{5}$	$\overline{16}$	$\overline{9}$	$\overline{11}$	$\overline{54}$	

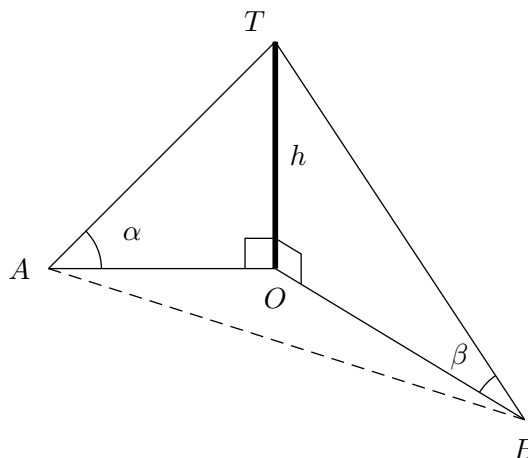
Question 1	(7 Marks)	Commence a NEW writing booklet	Marks
(a)	Find the exact value of $\tan 600^\circ$.		2
(b)	Solve for θ , if $\sin \theta = -\frac{\sqrt{3}}{2}$ and $0^\circ \leq \theta \leq 360^\circ$.		2
(c)	Solve for x , given $2 \cos^2 x - \cos x = 0$ and $0^\circ \leq x \leq 180^\circ$.		3

Question 2	(6 Marks)	Commence a NEW writing booklet	Marks
Let $f(x) = \sqrt{5 - 2x}$.			
(a)	Find the domain and range of $f(x)$.		2
(b)	Find the equation of the inverse function, $f^{-1}(x)$, and state its domain.		3
(c)	Briefly explain why $f(f^{-1}(x)) \neq x$ where $x < 0$.		1

Question 3	(5 Marks)	Commence a NEW writing booklet	Marks
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A vertical flagpole OT of height h metres stands with its base O on horizontal ground. Additionally, A and B are points on the ground.

From A and B , the angles of elevation to the top T of the flagpole are α and β respectively, where $\tan \alpha = \frac{5}{3}$ and $\tan \beta = \frac{5}{4}$.



(a)	Find OA and OB in terms of h .	2
(b)	Given that $\angle AOB = 120^\circ$, find AB in terms of h as an exact value.	3

Examination continues overleaf...

Question 4 (16 Marks) Commence a NEW writing booklet **Marks**

(a) Use differentiation by first principles to find $f'(x)$ given $f(x) = -2x^2$. **2**

(b) Differentiate the following with respect to x . Leave your answers in the simplest form.

i. $x(2x + 1)^4$ **2**

ii. $\frac{2x^2 - 1}{x + 1}$ **2**

(c) Consider the curve

$$f(x) = 2x^3 - 7x^2 + 4x + 6$$

i. Find the coordinates of the turning points and determine their nature. **5**

ii. Find the coordinates of any point(s) of inflection. **2**

iii. Hence sketch the graph of $y = f(x)$, showing the stationary points, the point(s) of inflection and the y -intercept. **3**
(Do NOT find the x -intercept.)

Question 5 (9 Marks) Commence a NEW writing booklet **Marks**

(a) Consider the curve defined by

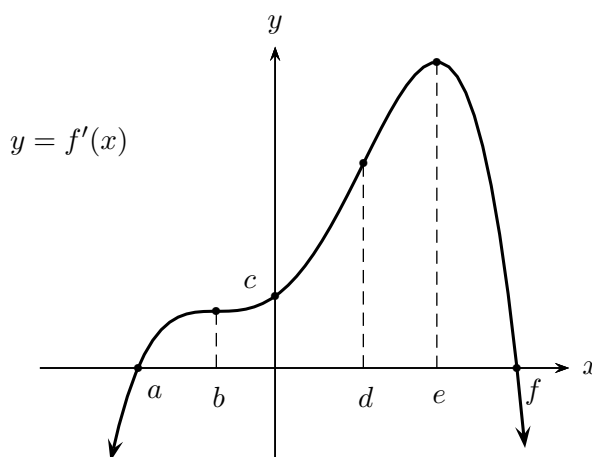
$$x = t^2 + 2t \quad \text{and} \quad y = t^{-3} - 1$$

where $t > 0$.

i. Find the expression for $\frac{dy}{dx}$, in terms of t . **2**

ii. Hence, find the equation of the tangent to the curve where $t = 1$. **3**

(b) The diagram below shows a sketch of $y = f'(x)$ with x -intercepts at $x = a$ and $x = f$, y -intercept at $y = c$, stationary points at $x = b$ and $x = e$, and a point of inflection at $x = d$.



i. State the value(s) of x for which the original function $y = f(x)$ has a stationary point. **2**

ii. Sketch the graph of $y = f''(x)$, showing all important features. **2**

Examination continues overleaf...

Question 6	(11 Marks)	Commence a NEW writing booklet	Marks
(a)	Fully simplify $3 \log_a \sqrt{a^5} + \log_a \frac{1}{a^2}$.		2
(b)	Given that $7^{-\frac{x}{2}} = \frac{1}{3}$ and $7^{\frac{y}{4}} = 2$, evaluate $\frac{1}{7^{y-x}}$.		3
(c)	i. Prove that $\log_{\frac{1}{x}} \frac{1}{y} = \log_x y$ where x and y are both positive numbers and $x \neq 1$.		2
	ii. Hence solve the equation $4 \log_x 3 + \log_3 27 = \log_{\frac{1}{3}} \frac{1}{x}$.		4

End of paper.

Question 1 (7 marks) (LEE)

(a) (2 marks)

- ✓ [1] for correct sign
- ✓ [1] for exact value

$$\begin{aligned}
 \tan 600^\circ &= \tan 240^\circ \\
 &= \tan 60^\circ \\
 &= \sqrt{3} \quad \checkmark \checkmark
 \end{aligned}$$

(b) (2 marks)

- ✓ [1] for related reference angle
- ✓ [1] for final answers

$$\begin{aligned}
 \sin \theta &= \frac{-\sqrt{3}}{2} \\
 \text{ref } \angle &= 60^\circ \quad \checkmark \\
 x &= 240^\circ, 300^\circ \quad \checkmark
 \end{aligned}$$

(c) (3 marks)

- ✓ [1] for factorising and/or obtaining two trigonometric equations
- ✓ [1] for solution to $\cos x = 0$
- ✓ [1] for solution to $\cos x = \frac{1}{2}$

$$\begin{aligned}
 \cos x(2 \cos x - 1) &= 0 \quad \checkmark \\
 \cos x &= 0 \quad \text{or} \quad \cos x = \frac{1}{2} \\
 \therefore x &= 60^\circ, 90^\circ \quad \checkmark \checkmark
 \end{aligned}$$

Question 2 (6 marks) (SEKARAN)

(a) (2 mark)

- ✓ [1] for domain
- ✓ [1] for range

$$\begin{aligned}
 \text{Domain : all real } x &\leq \frac{5}{2} \quad \text{or} \quad \left[-\infty, \frac{5}{2}\right) \quad \checkmark \\
 \text{Range : all real } y &\geq 0 \quad \text{or} \quad [0, \infty) \quad \checkmark
 \end{aligned}$$

(b) (3 marks)

✓ [1] for procedure of swapping x and y ✓ [1] for $f^{-1}(x)$ ✓ [1] for domain of $f^{-1}(x)$

$$\text{Let } y = \sqrt{5 - 2x}, \text{ where } x \leq \frac{5}{2}$$

$$\text{Then inverse has equation } x = \sqrt{5 - 2y}, \text{ where } y \leq \frac{5}{2} \quad \checkmark$$

$$y = \frac{5 - x^2}{2}$$

$$\therefore f^{-1}(x) = \frac{5 - x^2}{2} \quad \checkmark$$

$$\text{Domain : all real } x \geq 0 \quad \text{or} \quad [0, \infty) \quad \checkmark$$

(c) (1 mark)

✓ [1] for appropriate reasoning

Since the domain of $f^{-1}(x)$ is all real $x \geq 0$, when $x < 0, f(f^{-1}(x)) \neq x \quad \checkmark$

Question 3 (5 marks) (HO)

(a) (2 marks)

✓ [1] for OA in terms of h ✓ [1] for OB in terms of h

$$\tan \alpha = \frac{h}{OA}$$

$$\frac{5}{3} = \frac{h}{OA}$$

$$\therefore OA = \frac{3h}{5} \quad \checkmark$$

$$\tan \beta = \frac{h}{OB}$$

$$\frac{5}{4} = \frac{h}{OB}$$

$$\therefore OB = \frac{4h}{5} \quad \checkmark$$

(b) (3 marks)

- ✓ [1] for applying cosine rule
- ✓ [1] for simplification of trigonometric functions
- ✓ [1] for final answer

Using Cosine rule,

$$AB^2 = OA^2 + OB^2 - 2(OA)(OB) \cos 120^\circ \quad \checkmark$$

$$AB^2 = \left(\frac{3h}{5}\right)^2 + \left(\frac{4h}{5}\right)^2 - 2\left(\frac{3h}{5}\right)\left(\frac{4h}{5}\right) \cos 120^\circ$$

$$AB^2 = \frac{9h^2}{25} + \frac{16h^2}{25} + \frac{12h^2}{25} \quad \checkmark$$

$$AB^2 = \frac{37h^2}{25}$$

$$\therefore AB = \frac{\sqrt{37}h}{5} \quad \checkmark$$

Question 4 (16 marks) (J. KIM and LAM)

(a) (2 mark) (J. KIM)

- ✓ [1] for applying first principles and/or difference quotient
- ✓ [1] for final answer

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2(x+h)^2 - (-2x^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2x^2 - 4xh - 2h^2 + 2x^2}{h} \quad \checkmark \\ &= \lim_{h \rightarrow 0} -4x - 2h \\ &= -4x \quad \checkmark \end{aligned}$$

(b) i. (2 marks) (J. KIM)

- ✓ [1] for applying product rule
- ✓ [1] for final answer

$$\begin{aligned} \frac{d}{dx} (x(2x+1)^4) &= (2x+1)^4(1) + x(4(2x+1)^3 \times 2) \quad \checkmark \\ &= (2x+1)^3(10x+1) \quad \checkmark \end{aligned}$$

ii. (2 marks) (J. KIM)

✓ [1] for applying quotient rule

✓ [1] for final answer

$$\begin{aligned}\frac{d}{dx} \left(\frac{2x^2 - 1}{x + 1} \right) &= \frac{(x + 1)(4x) - (2x^2 - 1)(1)}{(x + 1)^2} && \checkmark \\ &= \frac{2x^2 + 4x + 1}{(x + 1)^2} && \checkmark\end{aligned}$$

(c) i. (5 marks) (LAM)

✓ [1] for the function of $f'(x)$

✓ [2] for coordinates of stationary points

✓ [2] for nature of stationary points

$$\begin{aligned}f'(x) &= 6x^2 - 14x + 4 && \checkmark \\ f''(x) &= 12x - 14\end{aligned}$$

Stationary points occur when $f'(x) = 0$:

$$\begin{aligned}(3x - 1)(x - 2) &= 0 \\ \therefore x &= \frac{1}{3} \quad \text{or} \quad x = 2\end{aligned}$$

$$\begin{aligned}f\left(\frac{1}{3}\right) &= \frac{179}{27} \\ f''\left(\frac{1}{3}\right) &= -10 < 0\end{aligned}$$

$\therefore \left(\frac{1}{3}, \frac{179}{27}\right)$ is a local maximum turning point. $\checkmark\checkmark$

$$\begin{aligned}f(2) &= 2 \\ f''(2) &= 10 > 0\end{aligned}$$

$\therefore (2, 2)$ is a local minimum turning point. $\checkmark\checkmark$

ii. (2 marks) (LAM)

✓ [1] for x -coordinate of point of inflection

✓ [1] for final answer with confirmation of change in concavity

Points of inflection occur when $f''(x) = 0$:

$$\begin{aligned}12x - 14 &= 0 \\ \therefore x &= \frac{7}{6} \\ f\left(\frac{7}{6}\right) &= \frac{233}{54}\end{aligned}$$

x	1	$\frac{7}{6}$	2
$f''(x)$	-2	0	10

There is a change in concavity. ✓

∴ $\left(\frac{7}{6}, \frac{233}{54}\right)$ is a point of inflection. ✓

iii. (3 marks) (LAM)

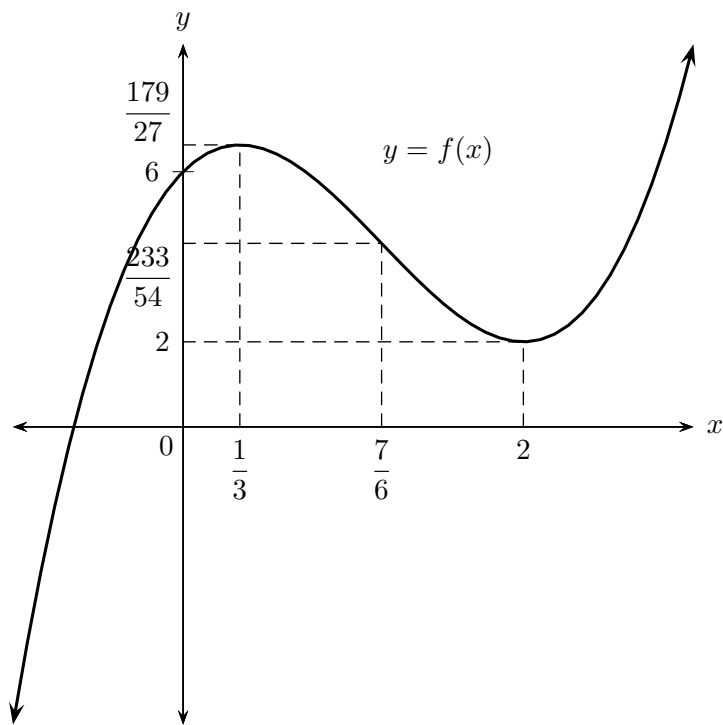
✓ [1] for shape and y -intercept

✓ [1] for stationary points

✓ [1] for point of inflection

$$f(0) = 6$$

∴ y -intercept at $(0, 6)$



Question 5 (9 marks) (HAM)

(a) i. (2 marks)

- ✓ [1] for $\frac{dy}{dt}$ and $\frac{dx}{dt}$
- ✓ [1] for final answer

$$\begin{aligned}\frac{dx}{dt} &= 2t + 2 \\ \therefore \frac{dt}{dx} &= \frac{1}{2t + 2} \\ \frac{dy}{dt} &= -3t^{-4} \\ &= \frac{-3}{t^4}\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{-3}{t^4} \times \frac{1}{2t + 2} \quad \checkmark \\ \therefore \frac{dy}{dx} &= \frac{-3}{t^4(2t + 2)} \quad \checkmark\end{aligned}$$

ii. (3 marks)

- ✓ [1] for x and y -coordinates when $t = 1$
- ✓ [1] for gradient of tangent
- ✓ [1] for final answer

When $t = 1$,

$$\begin{aligned}x &= 3 \\ y &= 0 \quad \checkmark \\ \frac{dy}{dx} &= \frac{-3}{4} \quad \checkmark\end{aligned}$$

Equation of tangent:

$$\begin{aligned}y - 0 &= -\frac{3}{4}(x - 3) \\ \therefore y &= -\frac{3}{4}x + \frac{9}{4} \quad \checkmark\end{aligned}$$

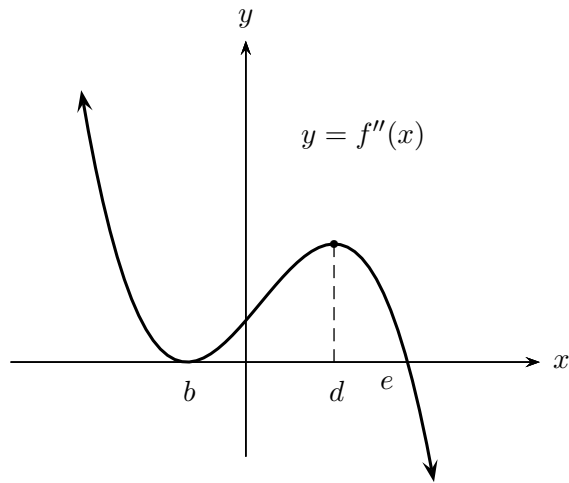
(b) i. (2 marks)

- ✓ [2] for final answers

At stationary point, $f'(x)$ crosses the x -axis. $\therefore x = a$ and $x = f$ ✓✓

ii. (2 marks)

- ✓ [1] for correct feature at $x = b$ or between $x = d$ and $x = e$
- ✓ [1] for final answer



Question 6 (11 marks) (CHAI)

(a) (2 marks)

- ✓ [1] for applying logarithmic laws
- ✓ [1] for final answer

$$\begin{aligned}
 3 \log_a \sqrt{a^5} + \log_a \frac{1}{a^2} &= 3 \log_a a^{2.5} + \log_a a^{-2} && \checkmark \\
 &= 3(2.5) - 2 \\
 &= 5.5 && \checkmark
 \end{aligned}$$

(b) (3 marks)

- ✓ [1] for evaluating 7^x and 7^y
- ✓ [1] for applying index laws to $\frac{1}{7^{y-x}}$
- ✓ [1] for final answer

$$\begin{aligned}
 7^{-\frac{x}{2}} &= \frac{1}{3} \implies 7^x = 9 \\
 7^{\frac{y}{4}} &= 2 \implies 7^y = 16 && \checkmark \\
 \frac{1}{7^{y-x}} &= 7^{x-y} \\
 &= \frac{7^x}{7^y} && \checkmark \\
 \therefore \frac{1}{7^{y-x}} &= \frac{9}{16} && \checkmark
 \end{aligned}$$

(c) i. (2 mark)

✓ [1] for applying logarithmic laws

✓ [1] for final answer

$$\begin{aligned}
 \text{LHS} &= \log_{\frac{1}{x}} \frac{1}{y} \\
 &= \frac{\log_x \frac{1}{y}}{\log_x \frac{1}{x}} \quad \checkmark \\
 &= \frac{-\log_x y}{-1} \quad \checkmark \\
 &= \log_x y \\
 &= \text{RHS}
 \end{aligned}$$

ii. (4 marks)

✓ [1] for applying part (i) results and other logarithm laws

✓ [1] for making $\log_3 x$ the subject of a quadratic equation

✓ [2] for final answers

$$\begin{aligned}
 4 \log_x 3 + \log_3 27 &= \log_{\frac{1}{3}} \frac{1}{x} \\
 \frac{4}{\log_3 x} + 3 &= \log_3 x \quad \checkmark \\
 4 + 3 \log_3 x - (\log_3 x)^2 &= 0 \quad \checkmark \\
 (4 - \log_3 x)(1 + \log_3 x) &= 0 \\
 \log_3 x &= 4 \quad \text{OR} \quad \log_3 x = -1 \\
 x &= 81 \quad \text{OR} \quad x = \frac{1}{3} \quad \checkmark \checkmark
 \end{aligned}$$